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TRANSMITTING DATA SIGNALS ON SEPARATE LAYERS OF A MEMORY MODULE, AND RELATED METHODS AND APPARATUSES

Abstract

Apparatuses and methods for routing and transmitting signals in an electronic device are described. An apparatus may include a printed circuit board (PCB) and circuitry. The circuitry may be configured to transmit a first set of data signals referenced to a first ground layer of the PCB and including data to be either read from or written to a single memory device. The circuitry may also be configured to transmit a second set of data signals referenced to a second, different ground layer of the PCB and including other data to be either read from or written to the single memory device.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/443,673, filed Jul. 27, 2021, which is a continuation of U.S. patent application Ser. No. 16/448,541, filed Jun. 21, 2019, now U.S. Pat. No. 11,107,507, issued Aug. 31, 2021, the disclosure of which is hereby incorporated herein in its entirety by this reference.

TECHNICAL FIELD

[0002] Embodiments of the disclosure relate to data signal routing (or transmitting), and more specifically, to routing data signals on separate layers of a memory module (rather than on the same layer). Yet more specifically, some embodiments relate to methods and apparatuses for such routing, and related memory modules, memory devices, semiconductor devices, and systems.

BACKGROUND

[0003] A dual in-line memory module (DIMM) is a module that contains multiple random access memory (RAM) chips on a substrate, which may be characterized as a circuit board. The DIMM circuit board includes a number of pins that connect the RAM chips and associated logic (i.e., memory controller) of the DIMM to a computer motherboard.

[0004] DIMMs are classified by ranks. A rank is an area or block of data (e.g., 64-bits). For example, a single-rank DIMM uses its dynamic RAM (DRAM) chips to create a single block of 64-bits, and each of the chips is activated by one chip-select (CS) signal from a memory controller. A dual-rank DIMM includes two blocks of data (e.g., two blocks of 64-bits) from two sets of chips on the DIMM, and uses two chip-select signals. In a dual-rank DIMM, a first block (e.g., rank 0) is located on a first side of the DIMM and a second block (e.g., rank 1) is located on an opposite, second side of the DIMM.

[0005] Conventional DIMM construction or design may not account for or attempt to avoid signal quality issues associated with transitions between layers of the DIMM device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a block diagram of a memory system including a number of memory devices, in accordance with various embodiments of the present disclosure.

[0007] FIG. 2 is a block diagram of a memory device, according to various embodiments of the present disclosure, in accordance with various embodiments of the present disclosure.

[0008] FIG. 3A is a block diagram of a conventional data signal routing scheme for a DIMM.

[0009] FIG. 3B is a block diagram of a data signal routing scheme for a memory module, in accordance with various embodiments of the present disclosure.

[0010] FIG. 4 is a flowchart illustrating an example method of transmitting data signals within a memory module, in accordance with various embodiments of the present disclosure.

[0011] FIG. 5 is a simplified block diagram of a semiconductor device, in accordance with various embodiments of the present disclosure.

[0012] FIG. 6 is a simplified block diagram of an electronic system, in accordance with various embodiments of the present disclosure.

DETAILED DESCRIPTION

[0013] As described above, in convention dual-rank DIMMs, data signals are routed (or transmitted) to a single memory chip along a single layer in the DIMM based on conventional data signal routing rules. For example, even DQ bits are routed along a layer of a dual-rank DIMM to a memory chip and odd DQ bits are routed along the same layer to the same memory chip. The data signals have different paths to the same memory chip, for example, even DQ bits are received at a top input layer of the DIMM and odd DQ bits are received at a bottom input layer of the DIMM. By routing the data signals (e.g., even/odd DQ bits) along the same layer in the DIMM, the data signals arrive at the memory chip at the same time (within picoseconds from one another).

However, such routing of the data signals along the same layer requires numerous reference changes that negatively affect the data signaling performance.

[0014] In various embodiments described herein, data signals (e.g., even/odd DQ bits) are routed to the same memory chip of a memory module (e.g., a dual-rank DIMM) along different layers of the memory module (e.g., different data signal (DQ) layers). For example, even DQ bits are routed along a first layer of the memory module and odd DQ bits are routed along a different, second layer of the memory module. Accordingly, the number of reference changes are reduced which improves the signal performance of the data signal.

[0015] Additionally, a memory controller may cause data signals that travel along different path (e.g., along different layers) to arrive at a memory chip at substantially the same time. For example, a memory controller may be trained to enable the data signals to travel along different path (e.g., along different layers) and arrive at the same memory chip at the same time. Alternatively, in one embodiment, the routing/transmitting is a function of a circuitry configuration of a DIMM and a memory controller interacts with the DIMM without concern for how the signals are transferred between the bus and the DRAMs. It should be appreciated that the terms “routing” and “transmitting,” as provided herein, are interchangeable. In general, the terms “routing” and “transmitting” describe the passing of data signals through one or more layers of a memory module (e.g., DIMM).

[0016] The term “reference change,” as provided herein, is defined as a change in reference from a first ground layer of a memory module (e.g., DIMM) to a different, second ground layer of the memory module. Such ground layers may also be referred to as ground planes and may be located within or may occupy a layer or layers or a printed circuit board (PCB). Layers of a PCB may include overlay, solder, signal, dielectric, ground, or the like, and each layer or each type of layer may be made of different materials or different combinations of materials relative to other layers or other types of layers.

[0017] FIG. 1 illustrates a memory system **100**, according to various embodiments of the present disclosure. Memory system **100** includes a number of memory devices **102**, **103**, **104** and **105** coupled to a communication bus **110** (e.g., a system bus). Each memory device **102-105** may include one or more memory die, and collectively, memory devices **102-105** may be referred to as a memory module (e.g., a dual in-line memory module (DIMM)), a multi-chip package (MCP) or a package on package (POP).

[0018] Memory system **100** further includes a controller **112** coupled to each memory device **102-105** via communication bus **110**. Controller **112**, which may include a processor or any other suitable type of controller, may be configured to control and/or regulate various operations of memory system **100**, as well as provide interactivity with another device or system coupled to memory system **100** via an interface **114**.

[0019] Communication bus **110** may include one or more of an address bus **120**, a data bus **122**,

and a control signal bus **124**. In some embodiments, memory devices **102-105**, communication bus **110**, and controller **112** may be configured (e.g., physically arranged and mounted) on a printed circuit board (PCB). In various embodiments, memory system **100** may include a DIMM and one or more memory devices **102-105** may be a rank (or a number of ranks) of the DIMM.

[0020] According to some embodiments of the present disclosure, at least some of memory devices **102-105** may be coupled to communication bus **110** via an associated interface **121A-121D** (referred to collectively as interface **121**). For example, interface **121** (any one of interface **121A-121D**) may include one or more nodes (e.g., input/output (I/O) nodes) for coupling signal lines of an associated memory device to respective signal lines of communication bus **110**. Further, interface **121** may include one or more nodes coupled to one or more power supplies (not shown in FIG. 1), such as, for example, power and/or reference potentials. For example, each interface **121** may include an electromechanical type connection or soldered lead connections to communication bus **110**.

[0021] A memory device (e.g., memory device **102**) may be in an active mode in response to the memory device being selected to drive data bus **122** to a particular state, such as in response to performing a read operation in the memory device. Further, the memory device (e.g., memory device **102**) may be in an inactive mode when another memory device (e.g., memory device **104**) is selected to drive data bus **122** to a particular state, such as in response to performing a read operation in the other memory device (e.g., memory device **104**).

[0022] FIG. 2 illustrates a memory device **200**, according to various embodiments of the present disclosure. Memory device **200**, which may include, for example, a DRAM (dynamic random access memory), a SRAM (static random access memory), a SDRAM (synchronous dynamic random access memory), a DDR SDRAM (double data rate DRAM, such as a DDR4 SDRAM and the like), or a SGRAM (synchronous graphics random access memory), may be part of memory system **100** of FIG. 1. For example, memory device **200** may include one of memory devices **102-105** of FIG. 1. Memory device **200**, which may be integrated on a semiconductor chip, may include a memory array **202**.

[0023] In the embodiment of FIG. 2, memory array **202** is shown as including eight memory banks BANK0-7. More or fewer banks may be included in memory array **202** of other embodiments. Each memory bank includes a number of word lines WL, a number of bit lines BL and/BL, and a number of memory cells MC arranged at intersections of the number of word lines WL and the number of bit lines BL and/BL. The selection of a word line WL may be performed by a row decoder **204** and the selection of the bit lines BL and/BL may be performed by a column decoder **206**. In the embodiment of FIG. 1, row decoder **204** may include a respective row decoder for each memory bank BANK0-7, and column decoder **206** may include a respective column decoder for each memory bank BANK0-7.

[0024] Bit lines BL and/BL are coupled to a respective sense amplifier SAMP. Read data from bit line BL or/BL may be amplified by sense amplifier SAMP, and transferred to read/write amplifiers **207** over complementary local data lines (LIOT/B), transfer gate (TG), and complementary main data lines (MIOT/B). Conversely, write data outputted from read/write amplifiers **207** may be transferred to sense amplifier SAMP over complementary main data lines MIOT/B, transfer gate TG, and complementary local data lines LIOT/B, and written in memory cell MC coupled to bit line BL or/BL.

[0025] Memory device **200** may be generally configured to be receive various inputs (e.g., from an external controller) via various terminals, such as address terminals **210**, command terminals **212**, clock terminals **214**, data terminals **216**, and data mask terminals **218**. Memory device **200** may include additional terminals such as power supply terminals **220** and **222**.

[0026] During a contemplated operation, one or more command signals COM, received via command terminals **212**, may be conveyed to a command decoder **250** via a command input circuit **252**. Command decoder **250** may include a circuit configured to generate various internal

commands via decoding one or more command signals COM. Examples of the internal commands include an active signal ACT and a read/write signal R/W.

[0027] Further, one or more address signals ADD, received via address terminals **210**, may be conveyed to an address decoder **230** via an address input circuit **232**. Address decoder **230** may be configured to supply a row address XADD to row decoder **204** and a column address YADD to column decoder **206**.

[0028] Active signal ACT may include a pulse signal that is activated in response to a command signal COM indicating row access (e.g., an active command). In response to active signal ACT, row decoder **204** of a specified bank address may be activated. As a result, the word line WL specified by row address XADD may be selected and activated.

[0029] Read/write signal R/W may include a pulse signal that is activated in response to a command signal COM indicating column access (e.g., a read command or a write command). In response to read/write signal R/W, column decoder **206** may be activated, and the bit line BL specified by column address YADD may be selected.

[0030] In response to active signal ACT, a read signal, a row address XADD, and a column address YADD, data may be read from memory cell MC specified by row address XADD and column address YADD. The read data may be output via a sense amplifier SAMP, a transfer gate TG, read/write amplifier **207**, an input/output circuit **262**, and data terminal **216**. Further, in response to active signal ACT, a write signal, a row address XADD, and a column address YADD, write data may be supplied to memory array **202** via data terminal **216**, input/output circuit **262**, read/write amplifier **207**, transfer gate TG, and sense amplifier SAMP. The write data may be written to memory cell MC specified by row address XADD and column address YADD.

[0031] Clock signals CK and /CK may be received via clock terminals **214**. A clock input circuit **270** may generate internal clock signals ICLK based on clock signals CK and /CK. Internal clock signals ICLK may be conveyed to various components of memory device **200**, such as command decoder **250** and an internal clock generator **272**. Internal clock generator **272** may generate internal clock signals LCLK, which may be conveyed to input/output circuit **262** (e.g., for controlling the operation timing of input/output circuit **262**). Further, data mask terminals **218** may receive one or more data mask signals DM. When data mask signal DM is activated, overwrite of corresponding data may be prohibited.

[0032] FIG. **3A** is a diagram of a conventional data signal routing scheme through a dual-rank DIMM **300A**. DIMM **300A** is a twelve layer DIMM. For example, DIMM **300A** includes layers L1-L12. Layer 1 (L1) is a top layer, layer 2 (L2) is a GND layer, layer 3 (L3) is a DQ/DQS layer, layer 4 (L4) is a CA/CTRL/CLK layer, layer 5 (L5) is a VDD plane layer, layer 6 (L6) is a CA/CTRL/CLK layer, layer 7 (L7) is a CA/CTRL/CLK layer, layer 8 (L8) is a VDD plane layer, layer 9 (L9) is a CA/CTRL/CLK layer, layer 10 (L10) is a DQ/DQS (Lower DRAMs) layer, layer 11 (L11) is a GND layer, and layer 12 (L12) is a bottom layer.

[0033] As described above, in a conventional data routing scheme (for example in a dual-rank DIMM), data signals are routed to a memory chip along a single layer in the DIMM. In a conventional data routing scheme, in response to a write operation, a memory controller (not shown) transmits data signals to top layer L1, and bottom layer L12 for routing to memory device **350** (e.g., rank 0 of a DDR4 DIMM). For example, data signals **310** (e.g., even DQ bits of data) are received at top layer L1 (e.g., a data input layer) and are routed through series resistor **315** (in layer L1) and then to layer L3 (e.g., a data signal (DQ) routing layer) by way of via **322**. Data signals **310** are then routed from layer L3 to memory device **350** by way of via **323**. In this example, data signals **310** do not experience a reference change because data signals **310** are referenced to layer L2 (GND layer) while being routed from layer L1 to memory device **350**.

[0034] Continuing with the example above, data signals **312** (e.g., odd DQ bits of the data) are received at bottom layer L12 (e.g., data input layer). Data signals **312** and data signals **310** are concurrently received at DIMM **300A**. Data signals **312** are routed through series resistor **317** (in

layer L12) and then to layer L3 by way of via **320**. Data signals **312** are then routed from layer L3 to memory device **350** through via **325**. In this example, data signals **312** experience one reference change because data signals **312** are initially referenced (e.g., upon receipt at DIMM **300A**) to a first GND layer (layer L11) and then change reference to a second GND layer (layer L2). It should be appreciated, that the term “concurrent” as described herein, in various embodiments, refers to signals arriving the same time (e.g., at a memory chip), or about the same time due to, for example, latency.

[0035] In another example, in a conventional data routing scheme, in response to a write operation, a memory controller (not shown) routes data signals to memory device **352** (e.g., rank 1 of a DDR4 DIMM). In such an example, data signals **310** (e.g., even DQ bits of data) are received at top layer L1 and are routed through series resistor **315** (in layer L1) and then to layer L3 through via **322**. Data signals **310** are then routed from layer L3 to memory device **352** through via **323**. In this example, data signals **310** experience one reference change because data signals **310** are initially referenced (e.g., upon receipt at DIMM **300A**) to a first GND layer (layer L2) and then change reference to a second GND layer (layer L11) while being routed from layer L1 to memory device **352**.

[0036] Continuing with the example above, data signals **312** (e.g., odd DQ bits of the data) are received at bottom layer L12 (e.g., a data input layer). Data signals **312** and data signals **310** are concurrently received at DIMM **300A**. Data signals **312** are routed through series resistor **317** (in layer L12) and then to layer L3 through via **320**. Data signals **312** are then routed from layer L3 to memory device **352** through via **325**. In this example, data signals **312** experience two reference changes. Data signals **312** experience a first reference change from GND layer (layer L11) to GND layer (layer L2) (e.g., near series termination). Data signals **312** experience a second reference change from GND layer at layer L2 to GND layer at layer L11 (e.g., proximate a DRAM ball grid array-not shown) when data signals **312** are routed from layer L3 to memory device **352**.

[0037] FIG. **3B** is a diagram of a data signal routing scheme through a memory module **300B**, according to various embodiments described herein. As a non-limiting example, memory module **300B** may include a DIMM. A data signal routing scheme described with respect to memory module **300B** reduces the number of reference changes compared to conventional modules. In particular, data signals are routed on different layers (rather than the same layer). By reducing the number of reference changes, performance of the data signals are improved.

[0038] As depicted in FIG. **3B**, memory module **300B** is a twelve layer memory module (similar to DIMM **300A**). For example, memory module **300B** includes layers L1-L12. Layer 1 (L1) is a top layer, layer 2 (L2) is a GND layer, layer 3 (L3) is a DQ/DQS layer, layer 4 (L4) is a CA/CTRL/CLK layer, layer 5 (L5) is a VDD plane layer, layer 6 (L6) is a CA/CTRL/CLK layer, layer 7 (L7) is a CA/CTRL/CLK layer, layer 8 (L8) is a VDD plane layer, layer 9 (L9) is a CA/CTRL/CLK layer, layer 10 (L10) is a DQ/DQS layer, layer 11 (L11) is a GND layer, and layer 12 (L12) is a bottom layer. In various embodiments, DIMM **300B** can have more or less than 12 layers. For example, DIMM **300B** may be a 14 layer DIMM.

[0039] As depicted, memory module **300B** includes a dual-rank memory module (e.g., a dual-rank DIMM). For example, memory module **300B** may include a DDR4-SDRAM DIMM, where memory device **350** is a first rank (e.g., rank 0) and memory device **352** is a second rank (e.g., rank 1). It should be appreciated that memory module **300B** can be, but is not limited to, a single-rank DIMM or a quad-rank DIMM.

[0040] Still referring to FIG. **3B**, in response to a write operation, a memory controller (e.g., DDR4 memory controller-not shown) transmits data signals via top layer L1 and bottom layer L12 for routing to memory device **350** (e.g., rank 0 of a DDR4 DIMM). For example, data signals **310** (e.g., even DQ bits of data) are transmitted to and received at top layer L1. Data signals **310** are then routed through series resistor **315** (in layer L1) and then to layer L3 through via **322**. It is noted that, according to some embodiments, via **322** extends from layer L1 to layer L3 (rather than

from layer L1 to layer L12, as depicted in FIG. 3A). Data signals **310** are then routed from layer L3 to memory device **350** through via **323**. In this example, data signals **310** do not experience a reference change because data signals **310** are referenced to layer L2 (GND layer) while being routed from layer L1 to layer L3 and to memory device **350**.

[0041] Continuing with the example above, data signals **312** (e.g., odd DQ bits of the data) are received via bottom layer L12 (e.g., data input layer). Data signals **312** and data signals **310** are concurrently received at memory module **300B**. Data signals **312** are routed through series resistor **317** (in layer L12) and then to layer L10 through via **320**. It is noted that, according to some embodiments, via **320** in FIG. 3B extends from layer L12 to layer L10 (rather than from layer L12 to layer L1, as depicted in FIG. 3A). Data signals **312** are then routed from layer L10 to memory device **350** through via **325**. In this example, data signals **312** experience one reference change because data signals **312** are initially referenced (e.g., upon receipt at memory module **300B**) to a first GND layer (layer L11) and then change reference to a second GND layer (layer L2) when routed to memory device **350**.

[0042] In another example, in response to a write operation, a memory controller (not shown) routes data signals to memory device **352** (e.g., rank 1 of a DDR4 DIMM). In such an example, data signals **310** (e.g., even DQ bits of data) are received at top layer L1 and are routed through series resistor **315** (in layer L1) and then to layer L3 through via **322**. Data signals **310** are then routed from layer L3 to memory device **352** through via **323**. In this example, data signals **310** experience one reference change from GND layer (layer L2) to GND layer (layer L11) while being routed from layer L1 to layer L3 to memory device **352**.

[0043] Continuing with the example above, data signals **312** (e.g., odd DQ bits of the data) are received at bottom layer L12. Data signals **312** and data signals **310** are concurrently received at memory module **300B**. Data signals **312** are routed through series resistor **317** (in layer L12) and then to layer L10 through via **320**. Data signals **312** are then routed from layer L10 to memory device **352** through via **325**. In this example, data signals **312** do not experience a reference change because data signals **312** remain referenced to GND layer (layer L11) when routed from layer L12 to layer L10 to memory device **352**. In one embodiment, GND layer at L11 and GND layer L2 are at common ground by way of GND via **321** and GND via **324**.

[0044] With reference again to DIMM **300A** of FIG. 3A, transmitting data from layer L1 to memory device **350** does not require a reference change, and transmitting data from layer L12 to memory device **350** requires one reference change. Further, transmitting data from layer L12 to memory device **352** requires one reference change, and transmitting data from layer L12 to memory device **352** requires two reference changes. In contrast, with reference to memory module **300B** of FIG. 3B, transmitting data from layer L1 to memory device **350** does not require a reference change, and transmitting data from layer L12 to memory device **350** requires one reference change. Further, transmitting data from layer L12 to memory device **352** requires one reference change, and transmitting data from layer L12 to memory device **352** does not require a reference change. Thus, compared to DIMM **300A**, a number of reference changes required in memory module **300B** is reduced, and thus signal quality may be increased.

[0045] Referring to at least FIG. 3B, a memory controller (e.g., memory controller **112**) controls, at least in part, routing of data signals through DIMM **300B**. A memory controller, in various embodiments, is a DDR4 memory controller. In various embodiments, the memory controller enables data signals **310** and **312** to be concurrently routed on different layers (e.g., layer L3 and layer L10) by eliminating timing differences between the data signals **310** and **312** as they are routed through different layers of the DIMM. It should be appreciated that the memory controller is able to eliminate the timing differences by training of the memory controller and/or implementing various routing algorithms.

[0046] In various embodiments, data signals **310** and data signals **312** are data signals for a byte/nibble of data. For example, data signals **310** are even DQ bits (e.g., D0, D2, D4) of a byte (or

nibble) of data and data signals **312** are odd DQ bits (e.g., D1, D3, D5) of the byte (or the nibble). Accordingly, data signals **310** can be referred to as a first subset (e.g., even DQ bits) of a byte/nibble of data, and data signals **312** can be referred to as a second subset (e.g., odd DQ bits) of the byte/nibble of data.

[0047] As described above, the routing schemes described with respect to FIGS. 3A and 3B are directed to writing data to DRAM. The routing schemes may also apply to read operations. The routing of read operations follow the opposite routing path as described for a respective write operation. For example, data signals **310** read from memory device **350** are routed on a reverse path of routing data signals **310** to memory device **350**.

[0048] Additionally, a number of reference changes for routing data signals during a read operation is the same number of reference changes during routing of a respective write operation. For example, as described above, data signals **310** experiences one reference change when routed to memory device **352**. Similarly, data signals **310** also experiences one reference change when routed from memory device **352** to layer L1 during a read operation.

[0049] Still referring to FIG. 3B, data signals **310** and **312** are routed on separate layers of DIMM **300B**. For example, data signals **310** are routed on layer L3 (either to memory device **350** or memory device **352**) while data signals **312** are routed on layer L10 (either to memory device **350** or memory device **352**). The layer of DIMM **300B** for routing of data signal **310** can be referred to as a top minus 2 layer (or top layer-2). That is, layer L3 is located two layers below top layer L1. In other words, the L3 layer is within two layers of top layer L1. Similarly, the layer of DIMM **300B** for routing of data signals **312** can be referred to as N minus 2 layer (or N-2 layer). That is, DIMM **300B** includes twelve layers (or N layers) and layer L10 is two layers above bottom layer L12. In other words, the L10 layer is within two layers of bottom layer L12. Accordingly, if DIMM **300B** was a 14 layer DIMM, data signals **310** would be routed on the top-2 layer (e.g., a third layer of the 14 layers) and data signals **312** would be routed on a N-2 layer (e.g., a twelfth layer of the 14 layers).

[0050] FIG. 4 is a flowchart of an example method **400** for transmitting (or routing) data signals through separate layers on a DIMM. Method **400** may be arranged in accordance with at least one embodiment described in the present disclosure. Method **400** may be performed, in some embodiments, by a device or system, such as memory system **100** of FIG. 1, DIMM **300B** of FIG. 3B, or another device or system. Although illustrated as discrete blocks, various blocks may be divided into additional blocks, combined into fewer blocks, or eliminated, depending on the desired implementation.

[0051] At block **410**, a first set of data signals associated with a memory access are transmitted on a first signal layer of a printed circuit board (PCB) of a dual-rank dual in-line memory module (DIMM) to or from a first memory device of the dual-rank DIMM. For example, referring to FIG. 3B, a memory controller transmits even data signal **310** (e.g., even DQ bits) of a byte of data on layer L3 (DQ/DQS layer) to memory device **350**. More specifically, a memory controller transmits the even data signal bits to memory device **350**.

[0052] At block **420**, a second set of data signals associated with the memory access are transmitted on a second signal layer of the PCB of the dual-rank DIMM to or from the first memory device of the dual-rank DIMM. For example, referring to FIG. 3B, a memory controller transmits odd data signal **312** (e.g., odd DQ bits) of a byte of data on layer L10 (DQ/DQS layer) to memory device **350**. More specifically, a memory controller transmits the odd data signal bits to memory device **350**.

[0053] In various embodiments, transmitting of data signals **310** on the first layer is concurrent with the transmitting of data signals **312** on the second layer. That is, a memory controller eliminates timing differences of data signals **310** and **312** during transmitting on separate respective layers to a single memory cell.

[0054] Modifications, additions, or omissions may be made to method **400** without departing from

the scope of the present disclosure. For example, the operations of method **400** may be implemented in differing order. Furthermore, the outlined operations and actions are only provided as examples, and some of the operations and actions may be optional, combined into fewer operations and actions, or expanded into additional operations and actions without detracting from the essence of the disclosed embodiment. For example, in various embodiments, a memory controller routes data signals on separate layers during a read operation.

[0055] A semiconductor device is also disclosed. The semiconductor device, which may include a memory device, may include one or more arrays (e.g., memory arrays). The semiconductor device may also include an output device including one or more output driver circuits, as described herein.

[0056] FIG. **5** is a simplified block diagram of a memory system **500** implemented according to one or more embodiments described herein. Memory system **500** includes a memory module including a number of memory devices **502** and a control logic component **504**. For example, memory devices **502** may include memory devices **102-105** of FIG. **1**, and/or memory device **350** and/or memory device **352** of FIG. **3B** and control logic component **504** may include controller **112** of FIG. **1**. Control logic component **504** may be operatively coupled with the memory devices **502** so as to read, write, or re-fresh any or all memory cells within memory devices **502**.

[0057] An electronic system is also disclosed. The electronic system may include memory system including a number of memory devices. FIG. **6** is a simplified block diagram of an electronic system **600** implemented according to one or more embodiments described herein. Electronic system **600** includes at least one input device **602**. Input device **602** may be a keyboard, a mouse, or a touch screen. Electronic system **600** further includes at least one output device **604**. Output device **604** may be a monitor, touch screen, or speaker. Input device **602** and output device **604** are not necessarily separable from one another. Electronic system **600** further includes a storage device **606**. Input device **602**, output device **604**, and storage device **606** are coupled to a processor **608**.

[0058] Electronic system **600** further includes a memory system **610** coupled to processor **608**. Memory system **610**, which may include memory system **100** of FIG. **1** or memory system **500** of FIG. **5**, includes a number of memory devices (e.g., memory device **102-105** of FIG. **1**). Electronic system **600** may include a computing, processing, industrial, or consumer product. For example, without limitation, electronic system **600** may include a personal computer or computer hardware component, a server or other networking hardware component, a handheld device, a tablet computer, an electronic notebook, a camera, a phone, a music player, a wireless device, a display, a chip set, a game, a vehicle, or other known systems.

[0059] According to various embodiments disclosed herein, and in contrast to some conventional methods, systems, and devices, data signals are routed along different layers of a memory module (e.g., a DIMM). As a result, reference changes for data signals are reduced. Accordingly, signal performance of the data signals is enhanced.

[0060] One or more embodiments of the present disclosure include an apparatus. The apparatus may include a printed circuit board (PCB) comprising a number of layers, a number of memory chips, and a circuitry coupled to the number of memory chips via one or more of the number of layers. The circuitry may be configured to transmit a first set of data signals associated with a memory access on a first signal layer of printed circuit board (PCB) to or from one of the number of memory chips. The circuitry may be configured to transmit a second set of data signals associated with the memory access on a second signal layer of the PCB to or from another one of the number of memory chips.

[0061] One or more embodiments of the present disclosure include an electronic system. The electronic system may include an input device, an output device, a processing device coupled to the input device and the output device, and a memory system coupled to the processing device. The memory system may include a first rank of memory chips, a second rank of memory chips, and circuitry coupled to the number of memory chips. The circuitry may be configured to transmit a first set of first data signals and a second set of the first data signals to or from the first rank of

memory chips without requiring more than one reference change for the first set of the first data signals and the second set of the first data signals. The circuitry may be configured to transmit a first set of second data signals and a second set of the second data signals to or from the second rank of memory chips without requiring more than one reference change for the first set of the second data signals and the second set of the second data signals.

[0062] One or more embodiments of the present disclosure include a method. The method may include transmitting a first set of data signals associated with a memory access on a first signal layer of a printed circuit board (PCB) of a dual-rank dual in-line memory module (DIMM) to or from a first memory device of the dual-rank DIMM. The method may also include transmitting a second set of data signals associated with the memory access on a second signal layer of the PCB of the dual-rank DIMM to or from the first memory device of the dual-rank DIMM.

[0063] In accordance with common practice, the various features illustrated in the drawings may not be drawn to scale. The illustrations presented in the present disclosure are not meant to be actual views of any particular apparatus (e.g., device, system, etc.) or method, but are merely idealized representations that are employed to describe various embodiments of the disclosure. Accordingly, the dimensions of the various features may be arbitrarily expanded or reduced for clarity. In addition, some of the drawings may be simplified for clarity. Thus, the drawings may not depict all of the components of a given apparatus (e.g., device) or all operations of a particular method.

[0064] Terms used herein and especially in the appended claims (e.g., bodies of the appended claims) are generally intended as “open” terms (e.g., the term “including” should be interpreted as “including, but not limited to,” the term “having” should be interpreted as “having at least,” the term “includes” should be interpreted as “includes, but is not limited to,” etc.).

[0065] Additionally, if a specific number of an introduced claim recitation is intended, such an intent will be explicitly recited in the claim, and in the absence of such recitation no such intent is present. For example, as an aid to understanding, the following appended claims may contain usage of the introductory phrases “at least one” and “one or more” to introduce claim recitations. However, the use of such phrases should not be construed to imply that the introduction of a claim recitation by the indefinite articles “a” or “an” limits any particular claim containing such introduced claim recitation to embodiments containing only one such recitation, even when the same claim includes the introductory phrases “one or more” or “at least one” and indefinite articles such as “a” or “an” (e.g., “a” and/or “an” should be interpreted to mean “at least one” or “one or more”); the same holds true for the use of definite articles used to introduce claim recitations. As used herein, “and/or” includes any and all combinations of one or more of the associated listed items.

[0066] In addition, even if a specific number of an introduced claim recitation is explicitly recited, it is understood that such recitation should be interpreted to mean at least the recited number (e.g., the bare recitation of “two recitations,” without other modifiers, means at least two recitations, or two or more recitations). Furthermore, in those instances where a convention analogous to “at least one of A, B, and C, etc.,” or “one or more of A, B, and C, etc.,” is used, in general such a construction is intended to include A alone, B alone, C alone, A and B together, A and C together, B and C together, or A, B, and C together, etc. For example, the use of the term “and/or” is intended to be construed in this manner.

[0067] Further, any disjunctive word or phrase presenting two or more alternative terms, whether in the description, claims, or drawings, should be understood to contemplate the possibilities of including one of the terms, either of the terms, or both terms. For example, the phrase “A or B” should be understood to include the possibilities of “A” or “B” or “A and B.”

[0068] Additionally, the use of the terms “first,” “second,” “third,” etc., are not necessarily used herein to connote a specific order or number of elements. Generally, the terms “first,” “second,” “third,” etc., are used to distinguish between different elements as generic identifiers. Absence a

showing that the terms “first,” “second,” “third,” etc., connote a specific order, these terms should not be understood to connote a specific order. Furthermore, absence a showing that the terms “first,” “second,” “third,” etc., connote a specific number of elements, these terms should not be understood to connote a specific number of elements.

[0069] The embodiments of the disclosure described above and illustrated in the accompanying drawings do not limit the scope of the disclosure, which is encompassed by the scope of the appended claims and their legal equivalents. Any equivalent embodiments are within the scope of this disclosure. Indeed, various modifications of the disclosure, in addition to those shown and described herein, such as alternative useful combinations of the elements described, will become apparent to those skilled in the art from the description. Such modifications and embodiments also fall within the scope of the appended claims and equivalents.

Claims

1. A method, comprising: transmitting a first set of data signals, referenced to a first ground layer of a printed circuit board (PCB), to or from a single memory device; and transmitting a second set of data signals, referenced to a second, different ground layer of the PCB, to or from the single memory device.
2. The method of claim 1, wherein transmitting the first set of data signals comprises transmitting the first set of data signal on a first signal layer adjacent the first ground layer.
3. The method of claim 1, wherein transmitting the second set of data signals comprises transmitting the second set of data signals on a second, different signal layer adjacent the second, different ground layer.
4. The method of claim 1, wherein: transmitting the first set of data signals comprises transmitting the first set of data signals with reference to a second layer of the PCB; and transmitting the second set of data signals on the second, different ground layer comprises transmitting the second set of data signals with reference to an eleventh layer of the PCB.
5. The method of claim 1, wherein: transmitting the first set of data signals comprises transmitting even DQ bits of a byte of data with reference to the first ground layer; and transmitting the second set of data signals comprises transmitting odd DQ bits of the byte of data with reference to the second, different ground layer.
6. The method of claim 1, wherein the transmitting the second set of data signals is concurrent with the transmitting the first set of data signals.
7. An apparatus, comprising: a printed circuit board (PCB); and circuitry configured to: transmit a first set of data signals referenced to a first ground layer of the PCB and including data to be either read from or written to a single memory device; and transmit a second set of data signals referenced to a second, different ground layer of the PCB and including other data to be either read from or written to the single memory device.
8. The apparatus of claim 7, wherein the first set of data signals are even DQ bits of a byte of data and the second set of data signals are odd DQ bits of the byte of data.
9. The apparatus of claim 7, wherein the first set of data signal are transmitted on a first layer of the PCB adjacent the first ground layer and the second set of data signals are transmitted on a second, different layer of the PCB adjacent the second, different ground layer.
10. The apparatus of claim 9, wherein the first layer is a third layer of the PCB and the second, different layer is a tenth layer of the PCB.
11. The apparatus of claim 7, wherein the first ground layer is adjacent a top layer of the PCB and the second, different ground layer is adjacent a bottom layer of the PCB.
12. The apparatus of claim 11, wherein the top layer is a first data input layer of the PCB and the bottom layer is a second data input layer of the PCB.
13. The apparatus of claim 7, wherein the first set of data signals are transmitted to or from the

single memory device while maintaining the reference to the first ground layer, and the second set of data signals are transmitted to or from the single memory device with no more than a single reference change.

14. The apparatus of claim 7, wherein the circuitry is further configured to: transmit even data bits of a byte of data with reference to the first ground layer; and concurrently transmit even data bits of the byte of data with reference to the second, different ground layer.

15. An apparatus, comprising: a first routing layer of a printed circuit board (PCB) configured to transmit a first set of a first number of data signals to or from a single memory device; a first ground layer of the PCB adjacent the first routing layer; a second, different routing layer of the PCB configured to transmit a second set of the first number of data signals; and a second ground layer of the PCB adjacent the second, different routing layer.

16. The apparatus of claim 15, further comprising a memory module comprising each of the first routing layer, the first ground layer, the second, different routing layer, and the second ground layer.

17. The apparatus of claim 15, wherein the first number of data signal include data to be either read from or written to the single memory device.

18. The apparatus of claim 17, wherein the first ground layer is adjacent a top layer of the PCB and second ground layer is adjacent a bottom layer of the PCB.

19. The apparatus of claim 15, wherein the first set of the first number of data signals are even data bits and the second set of the first number of data signals are odd data bits.

20. The apparatus of claim 15, wherein the first routing layer is within two layers of a top layer of the PCB and the second, different routing layer is within two layers of a bottom layer of the PCB.
